

Evaluation of Denoising Architectures on a Reconstructed Ultra-Low-Dose CT Dataset

Diogo L. Zago¹, Raíssa X. Contassot¹, Rafaela C. Ravazio¹,
Augusto O. Ughini¹, Lucas S. Kupssinski¹, Rodrigo C. Barros^{1,2}

¹Machine Learning Theory and Application Lab,
Pontifical Catholic University of Rio Grande do Sul,
90619-900 – Porto Alegre, RS, Brazil

²Kunumi Institute, Brazil

{d.zago002, raissa.contassot, augusto.ughini,
rafaela.ravazio}@edu.pucrs.br
{lucas.kupssinsku, rodrigo.barros}@pucrs.br

Abstract. *Advances in low-radiation-dose Computed Tomography bring the need to explore denoising methods. Recent techniques adopt Ultra-Low-Dose Computed Tomography (ULDCT) to improve patient safety; however, achieving diagnostically sufficient images requires removing a substantially larger amount of noise. This work evaluates different denoising methods using a ULDCT reconstructed dataset. The evaluated architectures range from convolutional networks to diffusion-based models. Among all evaluated models, CoreDiff achieved the best image quality, reaching an SSIM of 0.8385, a result also confirmed by visual inspection. Our results can serve as benchmarks for future studies on ULDCT denoising approaches.*

1. Introduction

Medical imaging plays an important role in disease diagnosis, and the most common exams rely on ionizing radiation, such as X-rays and Computed Tomography (CT) [Junda et al. 2021]. Compared with X-rays, CT provides a more detailed view of anatomical structures, capturing slice images of the body that can be reconstructed into a volume. This added detail comes with a cost, as the higher radiation doses may pose a long-term risk to patients [dos Santos Junior et al. 2020].

Low-Dose CT (LDCT) [Hu et al. 2021] and Ultra-Low-Dose CT (ULDCT) [Chen et al. 2026] are safer alternatives that reduce the radiation dose to as little as 5% of the normal dose, advancing the radiation protection goal while limiting patient exposure [Usui et al. 2021]. The literature reports analyses adopting different dose levels, particularly 10% or 5% of the normal dose [Jo et al. 2023]. However, the trade-off is that, by decreasing the radiation dose, both LDCT and ULDCT increase the image noise, which hinders radiological interpretation [Demir et al. 2023].

Denoising methods have been explored across different architectures. In 2015, U-Net [Ronneberger et al. 2015] was introduced as a convolutional encoder-decoder for image segmentation and has since been applied to a wide range of tasks. RED-CNN [Chen et al. 2017] is a pixel-wise convolutional encoder-decoder network designed for LDCT image denoising, which remains a reference to this day. Beyond these,

adversarial and perceptual approaches [Yang et al. 2018, Liang et al. 2020], attention-based and dynamic convolution models [Wang et al. 2023, Yang et al. 2023], and more recently diffusion-based methods [Gao et al. 2023, Li et al. 2023, Du et al. 2024, Gao et al. 2025], considered state-of-the-art in generative modeling, have also been investigated for LDCT denoising.

In this context, the objective of this work is to evaluate different deep learning architectures on the ULDCT denoising task using a reconstructed dataset. We assess how these architectures perform when denoising images acquired under ultra-low-dose protocols, which are characterized by severe noise levels.

2. Related Work

Early denoising approaches relied on deep convolutional networks. RED-CNN is a residual encoder-decoder that became a widely adopted benchmark [Chen et al. 2017], and U-Net introduced the skip-connection structure that serves as a reference for many later restoration methods [Ronneberger et al. 2015]. Refinements include ED-CNN [Liang et al. 2020], which combines dense connections with edge enhancement, and the dynamic convolution network of DPN [Yang et al. 2023]. These models are well validated at standard LDCT, but their behavior under the more aggressive ULDCT regime remains weakly explored.

Adversarial and attention-based approaches aim to improve perceptual quality. WGAN-VGG combines a GAN with Wasserstein distance and perceptual loss, achieving sharper reconstructions than MSE-supervised methods [Yang et al. 2018], and CT-former is a convolution-free dilated Vision Transformer that captures long-range dependencies [Wang et al. 2023]. The resulting trade-off between perceptual quality and structural fidelity has rarely been evaluated under ULDCT conditions, where the risk of hallucinated structures and loss of subtle anatomical detail is substantially higher.

Diffusion models have emerged as a promising alternative, balancing structural fidelity and perceptual quality. Du et al. proposed a structure-aware diffusion model (SAD) for LDCT [Du et al. 2024], Gao et al. presented CoreDiff [Gao et al. 2023] and a later noise-inspired variant aimed at cross-domain generalization [Gao et al. 2025], and FGDM explored zero-shot translation via frequency-guided diffusion [Li et al. 2023]. Most of these evaluations target only standard LDCT benchmarks. The present work addresses this gap by establishing a comparative baseline among representative convolutional, adversarial, Transformer-based, and diffusion-based architectures evaluated on ULDCT data, providing reproducible reference results for future research on CT translation under ultra-low-dose conditions.

3. Methodology

This work evaluates methods explored in the literature for the denoising task on an ULDCT reconstructed dataset.

3.1. Dataset

We used the 2016 Low-Dose CT Grand Challenge from the American Association of Physicists in Medicine (AAPM) and the Mayo Clinic [American Association of Physicists in Medicine 2016] dataset. The full dataset

comprises 30 patients, but only 10 have both Normal-dose CT (NDCT) and Low-dose CT (LDCT) available, with the latter corresponding to 25% of the normal ionizing radiation dose. We hold out patient L506 for testing, following the convention in the literature [Zhang et al. 2023, Wang et al. 2024, Zhao et al. 2024], and reserve patient L143 for validation [Sun et al. 2026], allocating the remaining eight patients for training.

3.1.1. Dataset Reconstruction for ULDCT

In this work, we explore the concept of ULDCT. The dataset originally provides LDCT images simulated at 25% of the normal radiation dose, but our goal is to study denoising under an ultra-low-dose condition. For this reason, instead of using the provided LDCT images, we started from the NDCT images, which correspond to the full radiation dose, and reconstructed new images simulating only 10% of that dose. The resulting ULDCT images therefore contain more noise than the standard LDCT images and allow us to evaluate how each architecture behaves under more severe noise levels. The reconstruction preserved resolution (512×512) and native pixel spacing (~ 0.665 mm). Our reconstruction is based on the approach in Structure-Aware Diffusion (SAD) [Du et al. 2024], which considers the fan-beam acquisition geometry. The parameters are listed in Table 1. Reconstruction was performed using ASTRA Toolbox [van Aarle et al. 2016, van Aarle et al. 2015, Palenstijn et al. 2011], with projection and back-projection accelerated by GPU.

Table 1. Reconstruction parameters used in the dataset.

Parameter	Value
Source-Detector Distance	1270.0 mm
Source-Origin Distance	870.0 mm
Origin-Detector Distance	400.0 mm
Number of Detectors	848
Detector Spacing	0.60 mm
Number of Views	720
Angular Increment	0.5° (360° cover)

For each slice of the NDCT datasets, six steps were followed: first, HU was converted to attenuation coefficient μ as per Equation 1, with $\mu_{water} = 0.02 \text{ mm}^{-1}$.

$$\mu = \left(\frac{HU}{1000} + 1 \right) \cdot \mu_{water} \quad (1)$$

Then, a clean sinogram was generated from the μ matrix. Filtered backprojection (FBP) was applied to the clean sinogram, generating a ground-truth image without noise. Noise injection followed Du et al. model [Du et al. 2024], considering $N_i \sim \text{Poisson}(I_0 \cdot e^{-p_i}) + \mathcal{N}(0, \sigma_e^2)$ and following Equation 2, where p_i is the attenuation value of the clean sinogram, I_0 controls the dose level by representing the intensity of incident photons and $\sigma_e^2 = 10$ is the variance of the electronic noise.

$$p_i^{noise} = -\ln\left(\frac{\max(N_1, 1)}{I_0}\right) \quad (2)$$

ULDCT was reconstructed using FBP on the noisy sinogram. Finally, μ was converted to HU, and the slices were normalized in the window [-1024, 3072] HU. The Parzen filter was chosen for reconstruction because of its ability to reproduce the official LDCT noise level in the AAPM-Mayo dataset. To obtain the noise relative to 10% of the normal dose, I_0 was set at 2×10^4 .

3.2. Architectures

3.2.1. Pre-processing

Training was performed using image patches rather than full 512×512 slices whenever required by the architecture or memory limitations. This strategy is commonly adopted in LDCT reconstruction studies because it increases the number of training samples, reduces GPU memory consumption, and allows the networks to focus on local anatomical and noise patterns. In this work, 64×64 patches were used as the standard training unit for patch-based models. For models that require full-resolution inference, the output image was reconstructed either directly from the full slice or by combining overlapping patches, depending on each method’s original inference protocol.

Data augmentation was applied only to the training set and included horizontal and vertical flips and rotations compatible with CT image geometry, applied consistently to the input ULDCT image and its paired NDCT target to preserve the supervised correspondence between both domains.

3.2.2. Evaluated Models

Ten architectures were evaluated in this study, based on their paradigm evolution: from pixel-wise convolutional networks, to adversarial and perceptual approaches, followed by attention-based and dynamic convolution models, and finally diffusion-based methods. This progression reflects the increasing emphasis of the literature on structural preservation, perceptual fidelity, and dose-level generalization. To ensure a controlled comparative analysis, all models were evaluated under the same dataset split, pre-processing pipeline, evaluation metrics, and computational environment. Model-specific optimization hyperparameters were kept as described by each original publication, since the compared methods belong to different paradigms and have substantially different training dynamics.

U-Net [Ronneberger et al. 2015]. A fully convolutional encoder-decoder originally proposed for biomedical image segmentation, with a contracting path, an expansive path, and skip connections between corresponding resolution levels. For CT reconstruction, the final classification layer was replaced by a single-channel regression output, preserving spatial resolution between input and output, converting the network into an image-to-image regression model trained to map ULDCT images into NDCT-like reconstructions.

RED-CNN [Chen et al. 2017]. A residual encoder-decoder designed specifically for LDCT denoising in the image domain. It uses symmetric convolutional and deconvolutional layers.

lutional layers connected by residual shortcuts, allowing the model to recover structural details while suppressing noise. We employed it without structural modifications for the ULDCT-NDCT mapping.

WGAN-VGG [Yang et al. 2018]. A generative adversarial framework based on the Wasserstein distance with gradient penalty, in which a discriminator encourages the generated image distribution to approximate the NDCT domain. It adds a perceptual loss computed from a pre-trained VGG-19. Since CT images are single-channel and VGG-19 expects three channels, the CT image was replicated across three channels before feature extraction.

EDCNN [Liang et al. 2020]. A densely connected convolutional network with an edge-enhancement module based on trainable Sobel-like filters, combining image restoration with explicit edge preservation. Its objective combines MSE with a multi-scale perceptual loss from a frozen ResNet-50, and the final output is added to the input through a residual connection. EDCNN was used in accordance with its original reconstruction formulation.

CTformer [Wang et al. 2023]. A convolution-free encoder-decoder transformer for LDCT denoising, using dilated Token2Token tokenization, cyclic shift operations, and Performer-like approximate attention to model long-range dependencies. During inference it operates on overlapping patches, discarding boundary regions and preserving only the central area of each patch to reduce border artifacts.

DPN/FRN [Yang et al. 2023]. A feature refinement network (FRN) combined with a dynamic perception network (DPN). The FRN uses dense residual blocks to refine low-level representations, while the DPN incorporates local channel attention and multi-scale dynamic dilated convolutions to preserve anatomical detail. The model was employed according to the original two-stage procedure, in which the FRN is pre-trained before end-to-end training.

CoreDiff [Gao et al. 2023]. A generalized diffusion model for LDCT reconstruction. Instead of defining the forward process as pure Gaussian noise corruption, it uses the LDCT image as the endpoint of the degradation process through a mean-preserving degradation operator. Its restoration network, CLEAR-Net, incorporates contextual and error-aware modulation, including information from adjacent slices, enabling reconstruction with fewer sampling steps.

FGDM [Li et al. 2023]. A frequency-guided diffusion model originally proposed for zero-shot image translation. It decomposes image information into low-frequency and high-frequency components, using low-pass filtering and Sobel-based high-frequency operators as conditional guidance, and generates intermediate frequency information with a denoising diffusion GAN backbone. We adapted FGDM to the supervised ULDCT-NDCT setting.

SAD [Du et al. 2024]. A structure-aware diffusion model based on Schrödinger Bridge theory. It introduces progressive self-guided structural prompts, extracted by a lightweight PiDiNet edge detector, together with an implicit conditional representation, preserving anatomical structures while enabling inference with one to five sampling steps.

NEED [Gao et al. 2025]. A noise-model-inspired diffusion framework operating

in two complementary domains, combining shifted Poisson diffusion (SPDiff) in the pre-log projection domain with double-guided Gaussian diffusion (DGDiff) in the image domain, integrated into a cascaded reconstruction strategy with adaptive timestep matching based on dose level.

Table 2. Summary of the evaluated models, paradigms, and main loss functions.

Model	Year	Paradigm	Main loss function	Trainable Parameters
U-Net	2015	CNN encoder-decoder	MSE regression loss	$\approx 31.0\text{M}$
RED-CNN	2017	Residual CNN autoencoder	MSE	$\approx 1.8\text{M}$
WGAN-VGG	2018	Wasserstein GAN	Adversarial + perceptual VGG-19 loss	$\approx 14.5\text{M}$
EDCNN	2020	CNN with edge enhancement	MSE + multi-scale perceptual loss	$\approx 80.9\text{k}$
CTformer	2023	Transformer	MSE	$\approx 1.4\text{M}$
DPN/FRN	2023	CNN with attention and dynamic convolution	L1	$\approx 1.5\text{M}$
CoreDiff	2023	Generalized diffusion	Two-stage restoration L2	$\approx 4.8\text{M}$
FGDM	2023	Frequency-guided diffusion GAN	Adversarial diffusion GAN loss	$\approx 67.2\text{M}$
SAD	2024	Schrödinger Bridge diffusion	Score matching	$\approx 14.9\text{M}$
NEED	2025	Dual-domain diffusion	Projection-domain + image-domain L2	$\approx 4.3\text{M}$

3.3. Training

In a comparative study, the fairness of the comparison depends on controlling the factors that are not the main object of analysis. Therefore, the dataset, patient-level train/validation/test split, pre-processing pipeline, evaluation metrics, and computational environment were kept consistent across all models. This ensures that the observed differences are primarily associated with the reconstruction method rather than data leakage, inconsistent normalization, or different evaluation conditions.

We kept the hyperparameters, including learning rate, batch size, optimizer, and number of epochs or iterations, according to the configuration of each original publication. Table 3 summarizes the training hyperparameters adopted for each architecture, and Table 2 summarizes the corresponding loss functions.

Table 3. Training hyperparameters adopted for each model.

Model	Learning rate	Batch size	Optimizer	Epochs
U-Net	1.0×10^{-2}	1	SGD with momentum	20
RED-CNN	1.0×10^{-4}	16	Adam	100
WGAN-VGG	1.0×10^{-6}	16	Adam (G+D, betas)	100
EDCNN	1.0×10^{-3}	32	AdamW	200
CTformer	1.0×10^{-5}	16	Adam	4000
DPN/FRN	1.0×10^{-4}	16	Adam	500 (+100 FRN)
CoreDiff	2.0×10^{-4}	4	Adam	10000
FGDM	1.0×10^{-4}	8	Adam (G+D, betas)	200
SAD	5.0×10^{-4}	4	Adam	10000
NEED	1.0×10^{-4}	8	Adam	10000

For DPN/FRN, the FRN subnetwork was pre-trained for 100 epochs before joint training of the complete architecture. For diffusion-based models, the number of iterations was considered the effective stopping criterion, while the number of epochs is reported

only as a nominal reference. When the original method specified a learning rate schedule, it was preserved. For example, DPN uses learning rate decay during training, while CTformer applies a gradual reduction of the learning rate down to 10^{-6} .

All models were implemented in PyTorch. During inference, diffusion models used accelerated few-step sampling according to their original proposals, such as ten steps for CoreDiff and one to five steps for SAD. This strategy was adopted to keep reconstruction time compatible with practical use while preserving the methodological identity of each diffusion approach.

3.3.1. Computational Environment

Training was initially conducted in the Kaggle environment, which is equipped with two NVIDIA Tesla T4 GPUs, each with 16 GB of memory. Due to the high computational cost of training, particularly for diffusion models requiring tens or hundreds of thousands of iterations, the experiments were later migrated to an NVIDIA RTX A6000 GPU with 48 GB of memory. The RTX A6000's larger memory capacity also enabled larger batches and full-resolution slices for models that required or benefited from this configuration.

3.4. Evaluation Metrics

We used three metrics that are standard in CT denoising studies: the Structural Similarity Index Measure (SSIM), the Peak Signal-to-Noise Ratio (PSNR), and the Root Mean Square Error (RMSE). Higher SSIM and PSNR indicate better reconstruction quality, while an RMSE closer to zero indicates higher precision.

SSIM evaluates the similarity between the reference and the processed image from a perceptual standpoint, combining luminance, contrast, and structure. It is defined in Equation 3 and computed in local windows to avoid inflated values, where μ_x and μ_y are the mean intensities of the images, σ_x^2 and σ_y^2 are the variances, σ_{xy} is the covariance, and C_1 and C_2 are constants for stability.

$$\text{SSIM} = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (3)$$

PSNR measures the quality of a reconstructed image by comparing the original signal to the noise that affects its fidelity after processing. It is defined in Equation 4, where MAX_I represents the maximum possible pixel value of the image.

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right) \quad (4)$$

RMSE measures the distance between the values predicted by a model and the real ones, and is reported here in Hounsfield Units. It is defined in Equation 5, where $\hat{y}_i$ is the predicted value, y_i is the real value, and n is the number of observations.

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{n}} \quad (5)$$

4. Results

Figure 1 shows the reconstructions for one abdominal slice of the test patient. CoreDiff produces the image closest to the target, with clean tissue boundaries and little residual noise, whereas U-Net yields the worst reconstruction and loses most of the anatomical content. The remaining models lie between these two extremes and are difficult to distinguish by eye, which motivates a quantitative comparison.

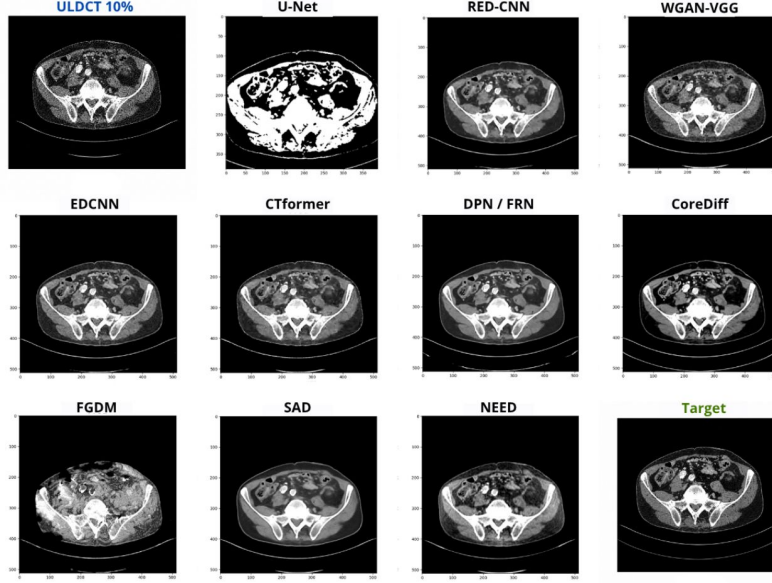


Figure 1. Comparison of the results on an abdominal slice for the test patient for each evaluated model.

Table 4 reports the mean metrics over all test slices. CoreDiff achieved the best values across the three metrics (SSIM 0.8385, PSNR 27.40 dB, RMSE 17.22 HU), followed by DPN/FRN. This shows that two different architectures, one based on diffusion and another on a combination of dynamic refinement with dynamic perception networks, performed well in a ULDCCT scenario.

The failure of U-Net deserves a separate comment, and it should be read as a training artifact rather than as an architectural finding. Table 3 shows that U-Net was the only model trained with SGD at a learning rate of 1.0×10^{-2} , with a batch size of 1 and only 20 epochs, against 100 to 10000 epochs for every other architecture. Under MSE supervision on high-variance ULDCCT inputs, this configuration is prone to divergence: the large gradient magnitudes produced by heavily corrupted targets, combined with single-sample batches and a comparatively high learning rate, yield unstable updates. The nearly binarized output in Figure 1, which does not preserve the Hounsfield intensity distribution of the reference, is consistent with a collapsed optimization rather than with an intrinsic limitation of the encoder-decoder. We therefore report U-Net as an under-trained control that marks the lower bound of the comparison, and we do not claim that segmentation-oriented architectures are unsuitable for ULDCCT denoising. Supporting such a claim would require re-training U-Net under a budget and an optimizer comparable to those of the remaining models.

A closer look at the results reveals that the diffusion paradigm alone does not

Table 4. Quantitative results of the evaluated models. Bold and underlined results represent the first and second-best values for each metric, respectively.

Model	SSIM [$\uparrow$]	PSNR (dB) [$\uparrow$]	RMSE (HU) [$\downarrow$]
U-Net*	0.4869	11.2591	109.8445
RED-CNN	0.7946	24.2491	24.6240
WGAN-VGG	0.7516	21.8028	32.5813
EDCNN	0.7963	24.1284	24.9627
CTformer	0.7804	23.3289	27.3368
DPN/FRN	<u>0.8103</u>	<u>25.3777</u>	<u>21.6592</u>
CoreDiff	<u>0.8385</u>	<u>27.4008</u>	<u>17.2163</u>
FGDM	0.6623	15.0843	70.8935
SAD	0.7803	23.6373	26.3983
NEED	0.7436	21.3967	34.1332

*Under-trained control, see discussion in Section 4.

explain performance. Four of the evaluated models are diffusion-based, namely CoreDiff, SAD, NEED, and FGDM, and they span almost the entire performance range. CoreDiff achieved the best results among all models, while FGDM and NEED were among the weakest. This indicates that success in ULDCT denoising depends on how each diffusion model is formulated for the task rather than on the diffusion framework itself.

In contrast, the convolutional models showed remarkable stability. RED-CNN, EDCNN, and DPN/FRN all fall within a narrow high-quality band, with DPN/FRN ranking second overall and first among non-diffusion models. It is worth noting that RED-CNN, a comparatively simple architecture proposed in 2017, remained competitive with the transformer-based CTformer and with more recent diffusion models. This suggests that in the ULDCT regime, architectural novelty yields diminishing returns, and that well-designed convolutional networks remain a reliable choice.

Figure 2 presents the ranking of the models related to image quality. For denoising ULDCT images, the model of choice should be CoreDiff. Other models, despite the differences in the architectures, behaved very similarly.

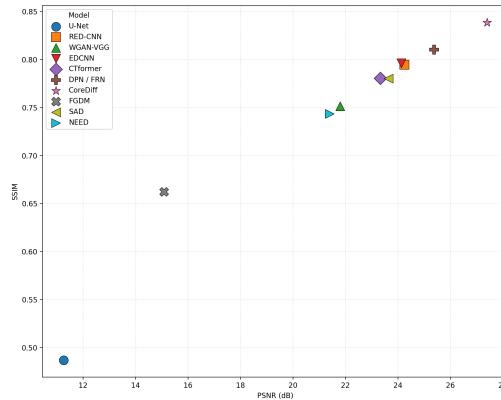


Figure 2. Comparison graph between SSIM and PSNR for all evaluated models.

Even the best-performing model reached around 27 dB in PSNR and an SSIM of 0.84, values that remain below those commonly reported for standard low-dose CT at 25% of the normal dose. This confirms that the 10% ultra-low-dose regime is substantially more challenging and that a considerable performance gap remains.

Our results define a baseline rather than a final ranking, and four limitations should be kept in mind. First, every model kept the hyperparameters of its original LDCT configuration, which supports a controlled comparison but penalizes methods that would benefit from re-tuning at 10% dose; U-Net is the clearest case. Second, the evaluation rests on a single test patient and a single seed, so we report no measure of variability across patients or runs. Third, SSIM, PSNR, and RMSE are pixel and structure based, and are known to penalize the perceptually oriented reconstructions favoured by GAN and diffusion models; perceptual metrics such as LPIPS or FID would give a fairer view of the generative methods. Fourth, the ULDCT simulation fixes $I_0 = 2 \times 10^4$ and $\sigma_e^2 = 10$, so the robustness of each architecture to other dose levels and electronic noise settings remains untested. Future work will re-train and re-tune the strongest models for the ultra-low-dose regime, extend the evaluation to additional test patients and seeds, add perceptual metrics together with per-slice inference time and peak memory, and vary the simulation parameters to probe robustness across scanner configurations.

5. Conclusion

Ultra-Low-Dose CT pushes radiation reduction further than conventional low-dose protocols, but the resulting increase in noise makes diagnostic interpretation considerably harder and turns denoising into a critical step. Motivated by this challenge, this work evaluated different denoising architectures on a reconstructed ULDCT dataset at 10% of the normal dose. We assessed 10 models, ranging from convolutional encoder-decoders such as U-Net and RED-CNN to transformer, adversarial, and diffusion-based approaches. CoreDiff achieved the best image quality, both qualitatively and quantitatively, leading in all evaluated metrics.

Our analysis also showed that the diffusion paradigm alone does not guarantee good performance, since diffusion models occupied both the top and the bottom of the ranking. Convolutional models proved to be the most reliable family, with RED-CNN, EDCNN, and DPN/FRN clustering in a high-quality band. The U-Net result, obtained under a substantially shorter training budget than the remaining models, is reported as an under-trained control rather than as evidence against segmentation-oriented architectures. As future work, we intend to investigate denoising methods designed specifically for the ultra-low-dose regime, to re-train the weakest configurations under comparable budgets, and to study how task-specific formulations can close the remaining performance gap at 10% of the dose.

Acknowledgments

This study was supported by the Center for Innovation and Artificial Intelligence in Health (CI-IA Saúde), with partial funding from the São Paulo Research Foundation (FAPESP), Grant No. 2020/09866-4; by the Minas Gerais Research Foundation (FAPEMIG), Grant No. PPE-00030-21; by UNIMED Belo Horizonte; by the Coordination for the Improvement of Higher Education Personnel (CAPES), Finance Code 001; by Conselho Nacional

de Desenvolvimento Científico e Tecnológico (CNPq), Grant No. 443072/2024-8; and by Kunumi Institute. The authors thank the institution for its financial support and commitment to advancing scientific research.

References

- American Association of Physicists in Medicine (2016). Low Dose CT Grand Challenge. <https://www.aapm.org/grandchallenge/lowdosect/>. Accessed: Apr. 24, 2026.
- Chen, H., Zhang, Y., Kalra, M. K., Lin, F., Chen, Y., Liao, P., Zhou, J., and Wang, G. (2017). Low-dose ct with a residual encoder-decoder convolutional neural network. *IEEE transactions on medical imaging*, 36(12):2524–2535.
- Chen, Z., Gao, Q., Li, Z., Zhang, J., Zhang, Y., Zhao, J., and Shan, H. (2026). Founddiff: Foundational diffusion model for generalizable low-dose ct denoising. *IEEE Transactions on Medical Imaging*.
- Demir, A. et al. (2023). Low-dose ct image enhancement using deep learning. *arXiv preprint arXiv:2310.20265*.
- dos Santos Junior, J. A., JansonNey, M. S. C., and de Souza Fonseca, G. V. (2020). Dose efetiva de radiação nos exames de tomografia computadorizada: um estudo retrospectivo e descritivo. *Diagnóstico e Tratamento*, 25(2):46–51.
- Du, W. et al. (2024). Structure-aware diffusion for low-dose ct imaging. *Physics in Medicine & Biology*, 69(15):155008.
- Gao, Q., Chen, Z., Zeng, D., Zhang, J., Ma, J., and Shan, H. (2025). Noise-inspired diffusion model for generalizable low-dose ct reconstruction. *Medical image analysis*, 105:103710.
- Gao, Q., Li, Z., Zhang, J., Zhang, Y., and Shan, H. (2023). Corediff: Contextual error-modulated generalized diffusion model for low-dose ct denoising and generalization. *IEEE Transactions on Medical Imaging*, 43(2):745–759.
- Hu, Q.-j. et al. (2021). Prospective study of low-and standard-dose chest ct for pulmonary nodule detection: a comparison of image quality, size measurements and radiation exposure. *Current Medical Science*, 41:966–973.
- Jo, G. D. et al. (2023). 75% radiation dose reduction using deep learning reconstruction on low-dose chest ct. *BMC Medical Imaging*, 23(1):121.
- Junda, M., Muller, H., and Friedrich-Nel, H. (2021). Local diagnostic reference levels for routine chest x-ray examinations at a public sector hospital in central south africa. *Health SA Gesondheid (Online)*, 26:1–8.
- Li, Y., Shao, H.-C., Liang, X., Chen, L., Li, R., Jiang, S., Wang, J., and Zhang, Y. (2023). Zero-shot medical image translation via frequency-guided diffusion models. *IEEE transactions on medical imaging*, 43(3):980–993.
- Liang, T., Jin, Y., Li, Y., and Wang, T. (2020). Edcnn: Edge enhancement-based densely connected network with compound loss for low-dose ct denoising. In *2020 15th IEEE International conference on signal processing (ICSP)*, volume 1, pages 193–198. IEEE.

- Palenstijn, W. J., Batenburg, K. J., and Sijbers, J. (2011). Performance improvements for iterative electron tomography reconstruction using graphics processing units (gpus). *Journal of Structural Biology*, 176(2):250–253.
- Ronneberger, O., Fischer, P., and Brox, T. (2015). U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pages 234–241. Springer.
- Sun, Y., Schneider, L.-S., Mei, S., Wang, J., Hu, G., Gu, M., Ye, C., Wagner, F., Song, L., Bayer, S., et al. (2026). Filter2noise: a framework for interpretable and zero-shot low-dose ct image denoising. *Journal of Medical Imaging*, 13(2):024004.
- Usui, K., Ogawa, K., Goto, M., Sakano, Y., Kyougoku, S., and Daida, H. (2021). Quantitative evaluation of deep convolutional neural network-based image denoising for low-dose computed tomography. *Visual Computing for Industry, Biomedicine, and Art*, 4(1):21.
- van Aarle, W., Palenstijn, W. J., Cant, J., Janssens, E., Bleichrodt, F., Dabrovolski, A., De Beenhouwer, J., Batenburg, K. J., and Sijbers, J. (2016). Fast and flexible x-ray tomography using the astra toolbox. *Optics Express*, 24(22):25129–25147.
- van Aarle, W., Palenstijn, W. J., De Beenhouwer, J., Altantzis, T., Bals, S., Batenburg, K. J., and Sijbers, J. (2015). The astra toolbox: A platform for advanced algorithm development in electron tomography. *Ultramicroscopy*, 157:35–47.
- Wang, B., Deng, F., Jiang, P., Wang, S., Han, X., and Zhang, Z. (2024). Witunet: A u-shaped architecture integrating cnn and transformer for improved feature alignment and local information fusion. *Scientific Reports*, 14(1):25525.
- Wang, D., Fan, F., Wu, Z., Liu, R., Wang, F., and Yu, H. (2023). Ctformer: convolution-free token2token dilated vision transformer for low-dose ct denoising. *Physics in Medicine & Biology*, 68(6):065012.
- Yang, Q., Yan, P., Zhang, Y., Yu, H., Shi, Y., Mou, X., Kalra, M. K., Zhang, Y., Sun, L., and Wang, G. (2018). Low-dose ct image denoising using a generative adversarial network with wasserstein distance and perceptual loss. *IEEE transactions on medical imaging*, 37(6):1348–1357.
- Yang, S., Pu, Q., Lei, C., Zhang, Q., Jeon, S., and Yang, X. (2023). Low-dose ct denoising with a high-level feature refinement and dynamic convolution network. *Medical Physics*, 50(6):3597–3611.
- Zhang, S.-Y., Wang, Z.-X., Yang, H.-B., Chen, Y.-L., Li, Y., Pan, Q., Wang, H.-K., and Zhao, C.-X. (2023). Hformer: highly efficient vision transformer for low-dose ct denoising. *Nuclear Science and Techniques*, 34(4):61.
- Zhao, H., Gu, Y., Zhao, Z., Du, B., Xu, Y., and Yu, R. (2024). Wia-ld2nd: Wavelet-based image alignment for self-supervised low-dose ct denoising. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 764–774. Springer.